Ali Nikkhah

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Summary

Backend and systems engineer with production experience building high-concurrency async APIs, distributed streaming infrastructure, and edge-optimized ML serving pipelines. At Digikala, engineered the async FastAPI serving layer handling 5M+ daily queries with multi-tier Redis caching, vLLM PagedAttention, and Triton Inference Server. At Advanced Analytics Australia, delivered distributed video analytics infrastructure on AXIS edge hardware using Kafka, TensorRT, and Kubernetes. Full ownership of the DevOps lifecycle: Docker, Kubernetes, Terraform, ArgoCD, Prometheus, and Grafana in production.

Technical Skills

Languages: Python (Asyncio, Pydantic), Go, C++, SQL, Bash, TypeScript
Deep Learning & AI: PyTorch, TensorFlow, JAX, Hugging Face, LangGraph, LangChain, LlamaIndex, FAISS, HyDE, KAG
Vision & Audio: OpenCV, ViT, BLIP, T5, Whisper, wav2vec2.0, TensorRT, TensorFlow Lite
Data & Streaming: Spark, Trino, Airflow, Kafka, PeerDB, Elasticsearch, Parquet
MLOps & Infra: Docker, Kubernetes, Terraform, Ray, Triton, vLLM, MLflow, Prometheus, Grafana

Professional Experience

Digikala — Tehran, Iran
AI Engineering R&D Lead (Backend & Systems) — *Jan 2025 – Aug 2025*

- Engineered the async **FastAPI** serving infrastructure handling over **5 million daily queries**; deployed **vLLM** with PagedAttention and **Triton Inference Server** for LLM serving; implemented multi-tier **Redis** caching and async request batching to fully isolate model execution latency from client-facing response times.
- Designed a Modular Control Pipeline (MCP) decoupling text ingestion, chunking, retrieval, and synthesis for zero-downtime LLM hot-swapping; built stateful **LangGraph** multi-agent routing with cyclic graphs to manage multi-turn session state and dynamic context evaluation.
- Supervised Kubernetes deployment of multi-tenant model serving clusters, isolating vision and language model compute nodes for independent horizontal scaling and fault containment.

Advanced Analytics Australia — Remote Contract
Backend & Edge Systems Engineer — *Sep 2025 – Apr 2026*

- Built high-performance async Python streaming workers interfacing with AXIS camera hardware, routing continuous live video feeds into a distributed **Apache Kafka** ingestion broker for fault-tolerant delivery under network spikes.
- Compiled and quantized vision networks into **TensorRT** and **TF Lite** runtimes served via Triton, eliminating Python-level inference overhead on resource-constrained edge nodes; implemented SNR monitoring for stable detection across variable-lighting deployments.

Turquoise Digital (Firouzeh Digital) — Tehran, Iran
Senior Data Engineer — *Sep 2025 – Mar 2026*

- Developed async **FastAPI** backend supporting high-frequency portfolio projection requests and automated trading execution; decoupled core relational databases from ingestion load via multi-tier **Redis** caching and **Apache Kafka** message queuing.
- Maintained full system observability across microservices with **Prometheus**, **Grafana**, and the ELK stack; orchestrated **Airflow** DAGs and deployed **PeerDB** CDC replication for sub-minute database mirroring into OLAP engines.

HomaCloud — Tehran, Iran
MLOps Engineer — *Aug 2021 – Feb 2022*

- Automated end-to-end model deployment with Docker, **ArgoCD**, and GitHub Actions (sub-10-minute releases); managed Kubernetes cluster orchestration and IaC via Terraform; built low-latency FastAPI inference APIs backed by TensorFlow Serving, scaling to concurrent traffic spikes.

Education

Sharif University of Technology — Tehran, Iran
B.Sc. Electrical Engineering – Digital Systems *GPA: 3.65 overall, 3.91 major* — *Sept 2020 – Jul 2024*

Research Highlights

Univ. of British Columbia *(Remote RA, Feb 2024 – Nov 2025)* — Vision-language architectures (ViT + BLIP + T5) with self-supervised contrastive learning for automated DICOM-compliant ultrasound report generation; achieved SOTA BLEU & ROUGE-L. Preprint in preparation.

L3S Research Center *(Remote RA, Apr 2023 – Feb 2024)* — 3D CNNs and transformer encoders over dense optical flow for texture-free motion classification; SOTA on UCF-101 and HMDB-51.